

Yunita Kristanti Andriani

Curriculum Vitae

She/Her

✉ andriani@het.phys.sci.osaka-u.ac.jp

Personal Information

Nationality Indonesian
Place/Date of Birth Malang (Indonesia)/July 28th, 2001

Education

April 2025 – **The University of Osaka**, Osaka, Japan.
present Department of Physics, *Particle Physics Theory Group*
July 2019 – **State University of Malang**, Malang, Indonesia.
May 2023 Department of Physics, *Computational and Theoretical Physics*
2016 – 2019 **State Senior High School 1 Tumpang**, Malang, Indonesia.
Department of Natural Sciences

Publications

- 2025 A. Salim Adam, **Y. K. Andriani**, and B. Dirgantara, Vacuum structure of an extended standard model with $U(1)_D$ symmetry, doi:[10.1007/JHEP01\(2026\)089](https://doi.org/10.1007/JHEP01(2026)089) [[arXiv:2509.05534](https://arxiv.org/abs/2509.05534) [[hep-ph](#)]].
- 2023 A. Salim Adam, **Y. K. Andriani**, and E. Latifah, Investigating Boson Sector in an Extended Standard Model with $U(1)_D$ Symmetry, doi:[10.1088/1402-4896/acfa49](https://doi.org/10.1088/1402-4896/acfa49) [[arXiv:2309.06852](https://arxiv.org/abs/2309.06852) [[hep-ph](#)]].

Talks, Posters, etc.

- March 2026 Talk entitled "Tree-Level Vacuum Stability Analysis of a Multi-Scalar $U(1)_D$ Extended Model" at New Higgs Working Group 42, held in Osaka, Japan
- February 2026 Talk entitled "Sphaleron in the Weinberg-Salam Theory" at M1 Journal Club, held by Particle Physics Theory Group of The University of Osaka, held in Osaka, Japan.
- June 2025 Poster entitled "Vacuum stability of an extended Higgs sector with multi-scalars under $U(1)$ global symmetries" at HPNP 2025, held in Osaka, Japan.

September 2023 Talk entitled "Investigation on Boson Sector and Vacuum Stability of a $U(1)_D$ Extended Standard Model" at Research Assistant Expo held by BRIN Research Center of Quantum Physics in Serpong City, Indonesia.

Experience

Research

May 2023 – Present **Vacuum Stability of an Extended Standard Model with Additional $U(1)_D$ Symmetry**

Supervisor: Apriadi Salim Adam, Sc.D. | Collaborator: Dr. Bayu Dirgantara

Work in progress

Abstract: In this research, we investigate the vacua of an extended standard model with $U(1)$ global symmetries, *i.e.* $U(1)_D \times U(1)_{B-L}$. The scalar sector consists of two $SU(2)$ doublets as well as one complex and one real singlets making the corresponding vacua slightly more complicated than the Standard Model one. We analyze theoretical constraints such as bounded-from-below conditions, global minimum, and perturbativity up to the Planck scale, alongside experimental bounds on the Higgs mass and its invisible decay. A detailed statistical analysis with numerical approach is performed, demonstrating that the extended scalar potential can accommodate a stable vacuum while satisfying both theoretical and experimental requirements for a small part of the parameter space.

December 2022 – April 2023 **The Mass of Scalar Higgs and Gauge Boson in an Extended Standard Model with Additional $U(1)_D$ Symmetry** – Bachelor Thesis

Advisor: Dr. Eny Latifah, S.Si., M.Si. | Supervisor: Apriadi Salim Adam, Sc.D.

This work was supported by the State University of Malang's 2023 Research Funding Program under Grant No. 5.4.1/UN32/KP/2023

Abstract: The insufficiency of the Standard Model (SM) of particle physics in explaining some of the observed phenomena has given rise to models with extended scalar sectors. In this study, we have investigated the gauge and Higgs sectors in an extended standard model (SM) with additional $U(1)_D$ symmetry. In the presented model, the singlet scalars and scalar doublets are added in addition to the SM-like scalar Higgs field. These additional scalars are also coupled to the gauge boson fields. The masses of both gauge and scalar Higgs bosons have also been calculated. Their masses are written in terms of the coupling and parameters of the potential, with their scale approximated by a brief numerical study.

Professional Training and Working

12 June 2024 – 25 June 2024 **2024 Asia Europe Pacific School of High-Energy Physics**, *Participating student*, Nakhon Pathom, Thailand.

- o Joined 31 lectures and 9 discussion sessions with high-energy physics students across the world.
- o Performed a collaborative project and presented the work in a group presentation.
- o see [school link](#) for more.

- 2 May 2023 – **Research Center for Quantum Physics – National Research and Innovation Agency (BRIN)**, *Research Assistant*, Southern Tangerang, Indonesia.
- 30 December 2023
- Engaged in research collaboration with my supervisor
 - Responsible for writing a monthly research report, research notes, and notes of discussions.
 - Responsible for conducting the research and reporting any progress made in each discussion session.
- September 2022 – **Research Center for Quantum Physics – National Research and Innovation Agency (BRIN)**, *Participant of Research Internship Program*, Southern Tangerang, Indonesia.
- February 2023
- Engaged in multiple intensive learnings related to high-energy physics and its experimental view
 - Involved in a research collaboration with one of the researchers (acted as a supervisor) from the research group of theoretical high-energy physics, about the mass generation of Higgs and gauge boson fields in an SM extension.
 - Communicated and discussed the learning process (and much more) with the advisors and other participants
 - Wrote a final report of general activities and the research that has been carried out
- 24 January – **Laboratory of Optics – State University of Malang**, *Laboratory and Teaching Assistant*, Malang, Indonesia.
- 20 May 2022
- Assisted 2 professors in managing experiments and class activities, including records (student attendance, experiment execution, assignment completion, and class schedule)
 - Supervised 2 different classroom discussions twice a week
 - Organized and maintained the laboratory equipment and supplies
 - Communicated and discussed the experiment with the Laboratory Manager and the team
 - Detail experiment: Single Slit Micro-diffraction Waves, Microwave: the Brewster angle effect, Michelson Interferometer, etc.
- August 2021 – **PT Microsoft Indonesia**, *Mentee of MBKM Internship Program*, Jakarta, Indonesia.
- January 2022
- Completed and achieved professional certifications in Microsoft Azure Data Fundamentals, Azure AI Fundamentals, and Power Platform Fundamentals
 - Worked on projects and discussions as an individual and team
 - Engaged in structured learning programs on Data Science and Artificial Intelligence
 - Carried out a final project by implementing learning machines and numerical data analysis
- November 2020 – **Non-institutional**, *Private Tutor*, Malang, Indonesia.
- July 2023
- Teach local elementary school, middle school, and high school students from ages 10 to 18 years old
 - Structure personalized courses, including Natural Science, Social Science, and English for all students
 - Provide offline and online forums for discussion and students' character improvement

Volunteer Work

- March 2024 – **Karang Taruna Desa Dawuhan**, *Volunteer Staff*, Malang, Indonesia.
May 2024 Joined for annual charity program.
- April 2023 – **Karang Taruna Desa Dawuhan**, *Volunteer Staff*, Malang, Indonesia.
June 2023 Joined for annual charity program.
- February 2021 **Karang Taruna Desa Dawuhan**, *Volunteer Staff*, Malang, Indonesia.
– June 2022
 - o Involved in multiple social events with a team, such as charity events where we distributed free groceries to local families in need
 - o Assisted the organization in multiple secretarial works, events organizing, etc.
- August – **Karang Taruna Permata Bangsa**, *Volunteer Staff*, Malang, Indonesia.
October 2020 Supervised local community and arranged a plan of outreach about the pandemic to local residents together with a team.

Awards, Grants, and Scholarships

- April 2025 – **Japanese Government (MEXT) Scholarship for Master's course** in The University of Osaka, Osaka, Japan.
Present
- September 2023 **Best presenter at Research Assistant (RA) EXPO 2023** held by the BRIN Research Center for Quantum Physics.
- July 2023 **Summa cum laude graduate**, Faculty of Mathematics and Natural Sciences, State University of Malang, with a cumulative GPA of 3.89 from 150 course credits.
- 2023 **Universitas Negeri Malang's 2023 Research Funding Program** for undergraduate final research project
- January 2020 **Outstanding physics freshman**, Faculty of Mathematics and Natural Sciences, State University of Malang, for obtaining 3rd highest GPA (3.85) in the first semester.

Technical skills

- Certified **Microsoft fundamentals** level in Azure Data, Azure AI, and Power Platform.
- Basic **L^AT_EX**
I have been working and familiar with L^AT_EX, especially in writing research notes, scientific papers, research posters, etc.
- Basic **Python**
I used to work with Python for machine learning training and data science. At the present moment, I use it for my research in complex statistical and numerical studies.
- Basic **Wolfram Mathematica**
I specifically use Mathematica for my research, particularly in noncomplex numerical calculations and simulations. In my recent work, I use RGBeta [[arXiv:2101.08265](#)] and SARAH [[arXiv:0806.0538](#)] in Mathematica.

Languages

Indonesian	<i>Native</i>
English	
Korean	<i>Basic</i>
Japanese	<i>Basic</i>

Miscellaneous

1. My hobbies particularly involved physical activities, such as martial arts (i.e. taekwondo since high school), walking, and jogging. I also draw a lot and used to submit several of my works to the annual magazine in high school.
2. Back in 2023, I was invited as one of the speakers at an expertise groups' introductory event in the Department of Physics of the State University of Malang. I shared my experience as a student from the theoretical and computational physics expertise group. The recorded event can be watched at the following [link](#) (talk given in Indonesian).

References

Prof. Shinya KANEMURA

Position: Professor

Affiliation: Dept. of Physics, The University of Osaka, Osaka, Japan

Email: kanemura(at)het.phys.sci.osaka-u.ac.jp